

Anup Sathya

PhD Student, University of Chicago

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anupsathya.com

Research

Anup's research explores how to protect the home from the encroachment of addictive platforms in the attention economy. Believing that learning to calmly coexist with smartphones is one of the defining challenges of the next century, he designs, builds, and studies physical devices that shield us from the constant digital onslaught.

Research Positions

Pre-Doctoral Researcher

2021 – 2022

Dept. of Computer Science, University of Maryland, College Park

Education

PhD, Computer Science

2022 – present

University of Chicago

Supervisor: Dr. Ken Nakagaki | Thesis: in progress

MS, Computer Science

2019 – 2021

University of Maryland, College Park

Supervisor: Dr. Huaishu Peng | Thesis: Enabling On-body Computing Using a Track Based Wearable

BTech, Electronics & Communication Engineering

2014 – 2018

PES University, Bangalore

Supervisor: Suresh Padmanabhan | Thesis: Realtime On-chip Wireless Waveform Monitoring

Major Grants & Funding

Arts, Science + Culture Graduate Collaboration Grant (\$3000)

2024

University of Chicago + School of the Art Institute of Chicago

Crerar Fellowship (\$5000)

2022

Dept. of Computer Science, University of Chicago

Daniels Fellowship (\$5000)

2022

Dept. of Computer Science, University of Chicago

Full tuition remission (~\$56k)

2019 – 2021

University of Maryland, College Park

Awards & Honors

Linda Tischler Award — Recognizing Emerging Design Talent Fast Company	2025
Innovation by Design Award (for Attention Receipts) Fast Company	2025
Lifestyle Accessories Award — Runner Up (for Attention Receipts) Core77	2025
Creative Hack Award Finalist (for Attention Receipts) WIRED Japan	2025
Special Recognition for Outstanding Reviews DIS 2023, CHI 2024	2024
Design Awards — Tools Award (for Fibercuit) Core77	2023
Nominated for Graduate Assistant of the Year (top 2%) University of Maryland, College Park	2020
Zonal Winner, National Robotics Championship IIT Bombay, India	2016

Publications

2026

- [17] ACM CHI '26 — Full Paper
Attention Nooks: Situated Frictions to Foster Intentional Technology Use
Anup Sathya, Ken Nakagaki
- [16] ACM TEI '26 — Full Paper
PopTuber: Discretely Reshaping High Resolution 3D Curves with "Serial Multistable" Tubes
Willa Yunqi Yang, Yifan Zou, Anup Sathya, Ken Nakagaki
Best Demo Award (People's Choice)
- [15] ACM TEI '26 — Work-in-Progress
FocuShift: Actuated Intervention to Prevent Smartphone Overuse via Phone-Attached Shape-Changing Device
Harrison Dong, Anup Sathya, Ken Nakagaki
- [14] ACM TEI '26 — Studio
Attention-Preserving Tangible Interfaces: Using Physicality and Materiality to Preserve Attention in the Home
Anup Sathya, Ken Nakagaki

2025

- [13] ACM UIST '25 — Full Paper
Shape n' Swarm: Hands-on, Shape-aware Generative Authoring with Swarm UI and LLMs
Matthew Jeung, Anup Sathya, Wanli Qian, Steven Arellano, Luke Jimenez, Ken Nakagaki
Best Demo Award
- [12] ACM UIST '25 — Workshop
Facilitating Longitudinal Interaction Studies of AI Systems

Tianyi Long, Sihan Wang, Émilie Fabre, Tingying Wang, *Anup Sathya*, Jing Wu, Savvas D. Petridis, Danqing Li, others

[11] ACM DIS '25 — Full Paper

Cybernetic Marionette: Channeling Collective Agency Through a Wearable Robot in a Live Dancer-Robot Duet

Anup Sathya, Jiasheng Li, Zeyu Yan, Adriane Fang, Bill Kules, Jonathan Martin, Huaishu Peng

2024

[10] ACM UIST '24 — Full Paper

SHAPE-IT: Exploring Text-to-Shape-Display for Generative Shape-Changing Behaviors with LLMs

Wanli Qian, Chen Gao, *Anup Sathya*, Ryo Suzuki, Ken Nakagaki

[9] ACM UIST '24 — Full Paper

CARDinality: Interactive Card-shaped Robots with Locomotion and Haptics using Vibration

Aditya Retnanto*, Emilio Faracci*, *Anup Sathya**, Yu Kai Hung, Ken Nakagaki [*equal contribution]

[8] ACM CHI '24 — Full Paper

Attention Receipts: Utilizing the Materiality of Receipts to Improve Screen-time Reflection on YouTube

Anup Sathya, Ken Nakagaki

FastCompany Innovation by Design Award, Core77 Lifestyle Accessories Award, WIRED Japan Creative Hack Award Finalist

[7] ACM UIST '24 — Poster

Game Jam with CARDinality: A Case Study of Exploring Play-based Interactive Applications

Emilio Faracci, Aditya Retnanto, *Anup Sathya*, Ashlyn Sparrow, Ken Nakagaki

2022

[6] ACM UIST '22 — Full Paper

Fibercuit: Prototyping High-Resolution Flexible and Kirigami Circuits with a Fiber Laser Engraver

Zeyu Yan*, *Anup Sathya**, Sarah Yusuf, Jyh-Ming Lien, Huaishu Peng [*equal contribution]

Best Demo Award (People's Choice), Core77 Tools Award - Notable

[5] ACM IMWUT (UBiCOMP) '22 — Full Paper

Calico: Relocatable On-cloth Wearables with Fast, Reliable, and Precise Locomotion

Anup Sathya, Jiasheng Li, Tauhidur Rahman, Ge Gao, Huaishu Peng

[4] ACM UIST '22 — Demo

Demonstration of Fibercuit: Techniques to Prototype High-Resolution Flexible and Kirigami Circuits with a Fiber Laser Engraver

Zeyu Yan*, *Anup Sathya**, Saadat Yusuf, Jyh-Ming Lien, Huaishu Peng [*equal contribution]

2021

[3] ACM CHI EA '21 — Poster

Towards On-the-wall Tangible Interaction: Using Walls as Interactive, Dynamic, and Responsive User Interface

Zeyu Yan, *Anup Sathya*, Pedro Jorge Carvalho, Yongquan Owen Hu, Ang Li, Huaishu Peng

2018

[2] IEEE ICACCI '18 — Short Paper

Realtime On-chip Wireless Waveform Monitoring

Anup Sathya, S. Balaji, Abhinav Gupta, S. Padmanabhan

2017

[1] IEEE TENCON '17 — Short Paper

Research Dissemination

Press

University of Chicago Magazine, 'Fit to print' — about <i>Attention Receipts</i>	2025
Hackster.io, 'A Game Changer' — about <i>CARDinality</i>	2024
inews UK, opinion piece citing <i>Attention Receipts</i>	2024
404 Media, 'Would You Waste Less of Your Life Online If You Got Daily Attention Receipts?'	2024
Adafruit Blog, 'Calico: A Wearable Robotic Assistant #WearableWednesday'	2023
Indian Express, 'This is Calico, a tiny robot that can be your dance instructor, workout tracker, and more'	2023
Communications of the ACM, 'A Wearable Robotic Assistant That's All Over You'	2023
IEEE Spectrum, 'A Wearable Robotic Assistant That's All Over You'	2023
The Verge, 'Wearable robot, why not?'	2023
Maryland Today, 'Where-able sensors'	2022
all3dp, Hackaday (×2), Hackster.io — coverage of Fibercuit and Calico	2022

Podcasts & Interviews

CBC Radio, 'As It Happens' — interviewed about <i>Attention Receipts</i>	2024
Put on Your Best...Robot, 'Over Coffee' — interviewed about <i>Calico</i>	2024
Hackster Café — interviewed with Huaishu Peng and Zeyu Yan about <i>Calico</i>	2022

Invited Talks

Intentional HCI — Designing for Intentional Technology Use	2025
University of British Columbia, Advanced Methods for Human-Computer Interaction, Dept. of Computer Science	

Teaching Experience

Teaching Assistant, A Practice in Art & Technology	2024
University of Chicago, Dept. of Computer Science Instructor: Ken Nakagaki	
Teaching Assistant, Intro to Human-Computer Interaction	2023
University of Chicago, Dept. of Computer Science Instructor: Ken Nakagaki	
Teaching Assistant, Actuated User Interfaces and Technologies	2022
University of Chicago, Dept. of Computer Science Instructor: Ken Nakagaki	

Service

Web Chair	2026
ACM TEI 2026	
Organizer, People and Tech Seminar	2024
Dept. of Computer Science, University of Chicago 2 quarters	
Reviewer	2022 – present
DIS 2023, CHI 2023, DIS 2024, CHI 2024, CHI 2025	